

SOLUTION FOR QUIZ-01 of GPE201 (WS2023-24)

Q.1

The seismic reflection method is sensitive to acoustic impedance contrast between two geological layers while the magnetic method is sensitive to magnetic susceptibility contrast. The magnetic method is a natural source field method while the seismic method can use either an artificial or a natural source.

Collecting seismic measurements requires contact with the ground while collecting magnetic data can be done at the surface or airborne.

Q.2

$$f = 1 - b/a$$

$$e^2 = 1 - b^2/a^2$$

$$\begin{aligned} \text{Therefore, } e^2 &= (1)^2 - (b/a)^2 \\ &= (1 - b/a) \times (1 + b/a) \\ &= f(1 + 1 - f) \\ e^2 &= f(2 - f) \end{aligned}$$

Q.3

The term "Great Oxidation Event" (also known as the Oxygen Catastrophe) occurred at about 2.5 - 2.3 Bya and refers to a significant geological period marked by a substantial increase in atmospheric oxygen level on the Earth. The GOE is inferred to have been caused by cyanobacteria that evolved porphyrin-based photosynthesis, which produced dioxygen as a byproduct.

Q.4

The mass of the Sun can be derived using the same methods that Newton employed with the Moon. Equating the force attracting the Earth to the Sun given by Newton's Law of Gravitation with that due to centripetal acceleration we get:

$$MmG/R^2 = mv^2/R \rightarrow M = v^2 R/G$$

where M is the mass of the Sun, m is the mass of the Earth, v is the velocity of the Earth around the Sun, G is the universal constant of gravitation and R is the distance of the Earth from the Sun.

Substituting $v = 2\pi R/P$ (where P is the period of the Earth's orbit) in the above equation, we get:

$$M = 4\pi^2 R^3 / GP^2 \quad \text{where } P = 365 \times 24 \times 60 \times 60 \text{ second, or, } P = 3.154 \times 10^7 \text{ second}$$

$$M = 4 \times (3.14159)^2 \times (1.496 \times 10^{11})^3 / (6.67 \times 10^{-11} \times (3.154 \times 10^7)^2) \text{ kg}$$

$$M = 1.99 \times 10^{30} \text{ kg}$$

Q.5

The magnetization of rock is related to the induced magnetic field H by the following formula:

$$M = kH, \text{ where } k \text{ is the susceptibility}$$

$$M = (2.8 \times 10^{-3}) \times (0.5 \times 10^4)$$

$$M = 14 \text{ A/m}$$

Q.6

Perihelion is the point in the orbital pathway of the Earth where it is closest to the sun.

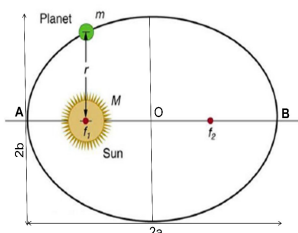
The closest distance (D_c) between the Sun and the Earth is given by the distance between points A and f_1 :

$$D_c = a - ae = a(1 - e),$$

where eccentricity (e) is given by $e = \sqrt{(a^2 - b^2)}/a$, **a** and **b** are semi-major and minor axes of the orbital pathway.

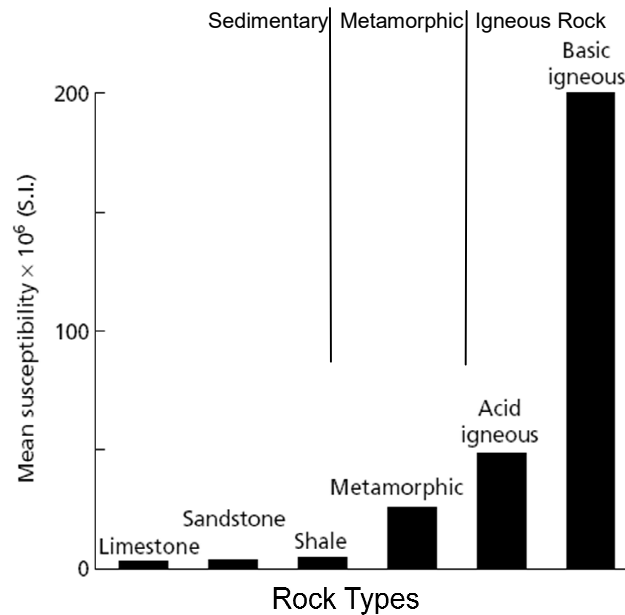
Aphelion on the other hand is the point in the orbit when the earth is farthest away from the sun.

The farthest distance (D_f) between Sun and the Earth is given by distance between points B and f_1 :
 $D_f = a + ae = a(1 + e)$



$OF_1 = OF_2 = ae$, where e is the eccentricity.
 Therefore distance between points A and f_1 is $a - ae = a(1 - e)$
 Similarly distance between points B and f_1 is $a + ae = a(1 + e)$

Q.7



Q.8

The mechanism accountable for the formation of the elements found within the periodic table is Nucleosynthesis, which refers to the process by which new atomic nuclei are formed through nuclear reactions, occurring primarily in stars and supernovae. The process makes heavier atoms by combining lighter atoms through nuclear fusion. This is where stars get the energy to shine. The two broad ways atomic nuclei form are by fusing smaller nuclei and nucleons (protons and neutrons) or by breaking apart larger nuclei through fission, radioactive decay, and other processes.

Q.9

An EON is the largest (formal) geochronologic time unit that contains the Archean, Proterozoic, Phanerozoic, and Hadean. The Hadean is an informal EON but is commonly used.

Q.10

The reflection coefficient, denoted as R , in a seismic survey is a measure of how much seismic energy is reflected at an interface between two geological layers. It helps us understand the contrast in properties (density and seismic velocity) between these layers and how much energy is transmitted or reflected when seismic waves encounter this boundary. To estimate the reflection coefficient (R) at the interface between the upper shale layer and the lower sandstone layer, the following formula can be used:

$$R = (\rho_2 V_2 - \rho_1 V_1) / (\rho_2 V_2 + \rho_1 V_1)$$

Where ρ_1, V_1 are density and seismic wave velocity in layer 1

Where ρ_2, V_2 are density and seismic wave velocity in layer 2

$$\text{Therefore } R = (2200 \times 2400 - 2300 \times 2950) / (2200 \times 2400 + 2300 \times 2950)$$

$$R = -0.13$$

Significance of negative and small value of R

- The negative sign of the reflection coefficient indicates that the reflected wave has an opposite polarity (phase) compared to the incident wave.
- The small magnitude of the reflection coefficient (-0.13) suggests that there is a very weak reflection at this interface. This implies that the two geological layers have relatively similar acoustic impedance.
- In this case, most of the energy is transmitted (the reflection is very weak), indicating that seismic waves can penetrate deeper into the lower sandstone layer.